Zero-Knowledge Multiplicative Non-Negative Proof (ZK-MNNP)

Pilar Smart City

Valda Orchidea Zahwa

Zero Knowledges Proof on Microlending

Overview

This research is a collaborative project between the Digital Trust Centre (NTU, Singapore) and ANT International (Alibaba Group). The initiative explores the use of Zero-Knowledge Range Proofs (ZKRP) to enhance data verification and credit assessment for e-commerce users applying for micro-loans or Buy Now Pay Later (BNPL) services.

Problem Context

Traditional micro-lending and paylater systems rely on centralized data access, forcing users to expose sensitive personal and financial information (e.g., total income, transaction history, spending behavior). This creates privacy and compliance challenges, especially under data protection laws such as PDPA and GDPR.

Zero-Knowledge Multiplicative Non-Negative Proof (ZK-MNNP)

Zero-Knowledge Multiplicative Non-Negative Proof (ZK-MNNP) is a cryptographic protocol designed to prove that a numerical value xxx lies within a valid range without revealing the actual value or its bounds. ZK-MNNP enables privacy-preserving verification in financial systems such as microlending and credit scoring. It allows multiple institutions (e.g., banks, fintechs, e-commerce platforms) to collaboratively verify borrower eligibility without exchanging raw financial data.

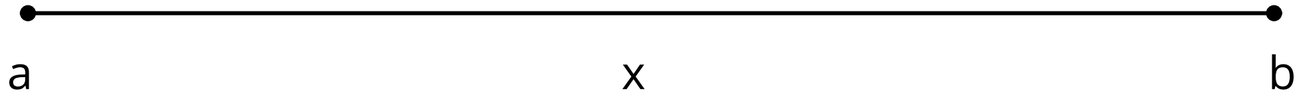
Hidden Bound Privacy

Hiding the lower and upper range limit verification example:

- Lender don’t know about amount income of borrower.
- Borrower don’t know minimum requirement to eligible for microlending.
- They only know that verify or not.

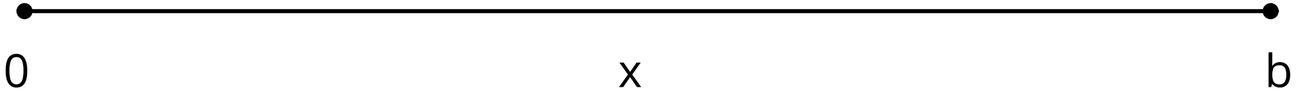
ZKRP

$a \leq x \leq b$



ZK Multi NNP

$x(B-x) \geq 0$



Nominal batas atas dan bawah di sembunyikan

Zero Knowledges Multiplicative Non Negative Proof

Multiconstraint paralleism

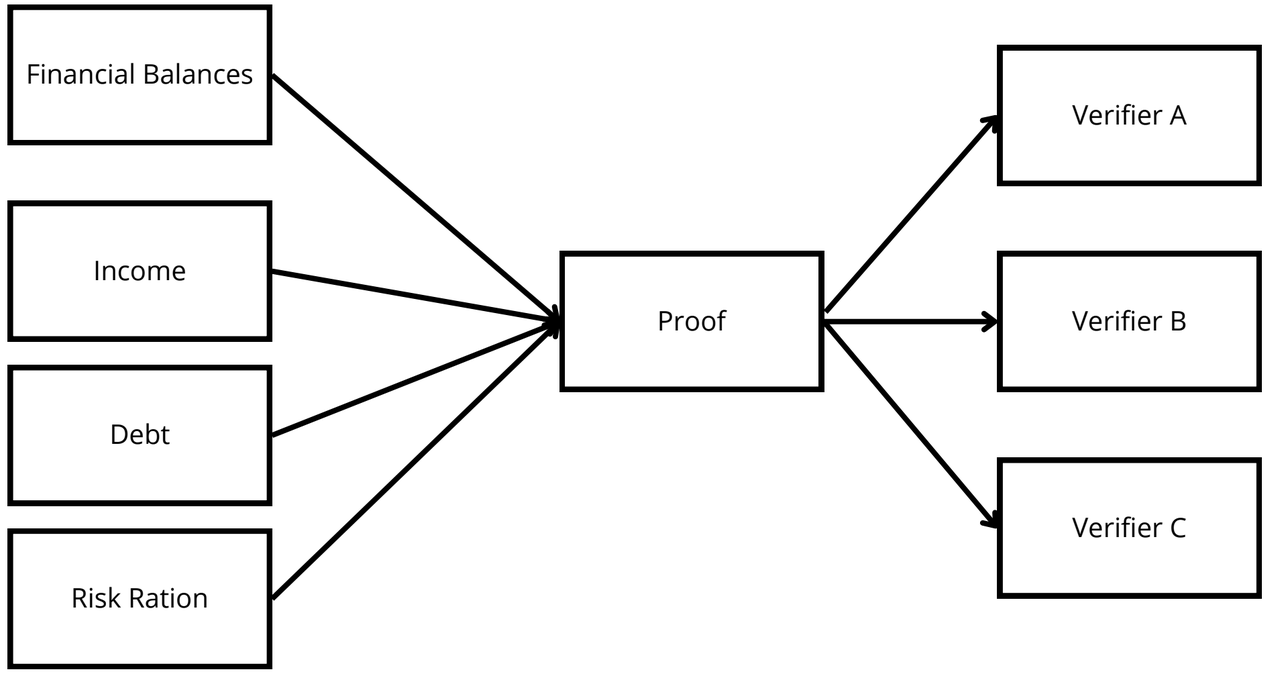
Generate proof with some constraint variable to be one proof. example Lender want to prove 3 variable, there are:

- Income > threshold
- debt < limit
- risk score < max allowed

Multiproof chaining

Verification technique from multiconstraint to be aggregate proof for verify the data for accross institution without revealing the data. Each verifier receives a partial view of the same proof transcript, enabling decentralized verification consistency

Multiconstraint



Multiproof chaining

Research Gap of Zero Knowledges Multiplicative Non Negative Proof

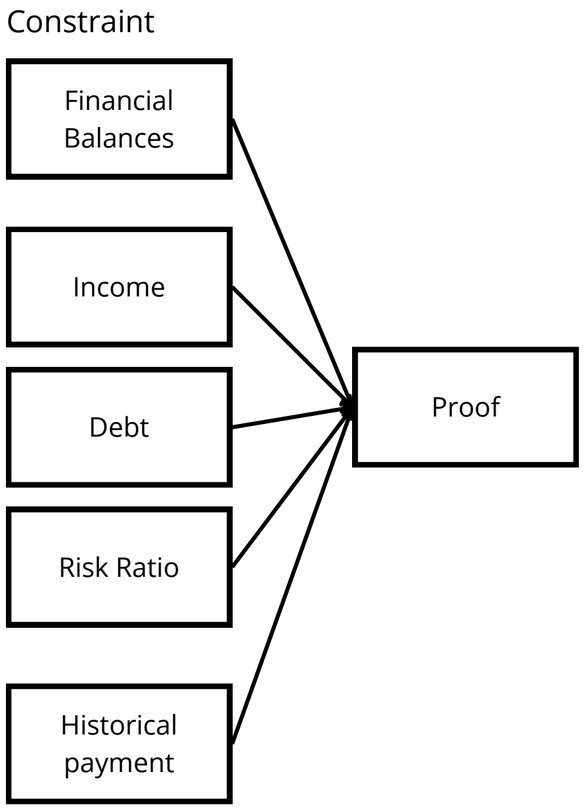
ZK-Multi NNP is not yet mature enough for real-world implementation. To verify an eligible borrower in microlending scenarios, data must often be collected from multiple institutions However, the current version of ZK-Multi NNP still relies on a single prover model, where all constraints are derived from a single data source and compiled into one proof. This approach limits scalability and privacy when multiple institutions need to collaborate without sharing raw data.

To address this limitation, ZK-Multi NNP needs to be extended into a multi-prover framework, where each institution can independently generate a local proof based on its own data (e.g., income, debt, or payment history).These local proofs can then be securely aggregated into a final global proof that can be verified by lenders or regulators without revealing sensitive information.

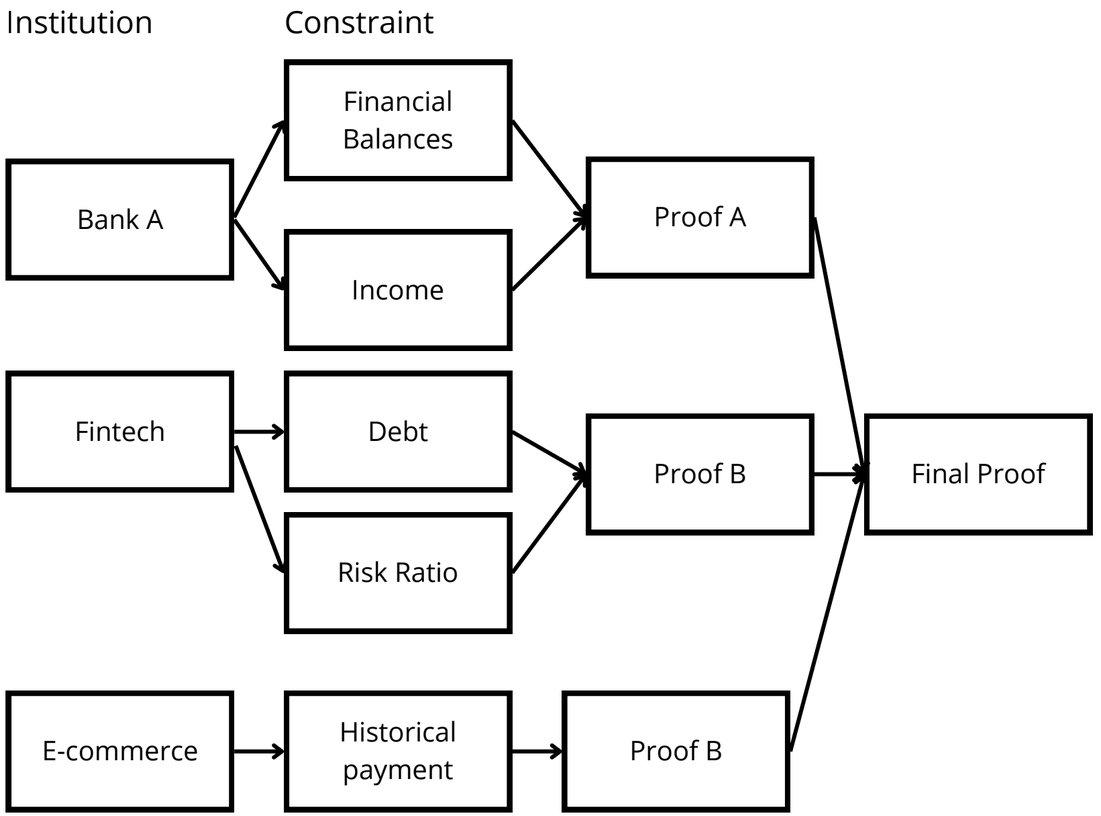
Concept	**Current Status**	**Proposed Improvement**
Proof Generation	Uses a single prover that generates one proof from all constraints in a centralized manner.	Each institution acts as an independent prover, generating partial proofs from its own
Data Source	Single-source data (monolithic).	Multi-source data from different trusted entities (banks, fintech, e-commerce).
Privacy & Scalability	Limited privacy — all data must be exposed to one prover.	Strong privacy — no institution needs to share raw data; only local proofs are exchanged.
Final Proof	Single proof = single data source.	Aggregated final proof = combination of multiple subproofs ($\pi_1, \pi_2, \ldots \pi_n$).

Research Gap of Zero Knowledges Multiplicative Non Negative Proof

Single prover



Multiprover

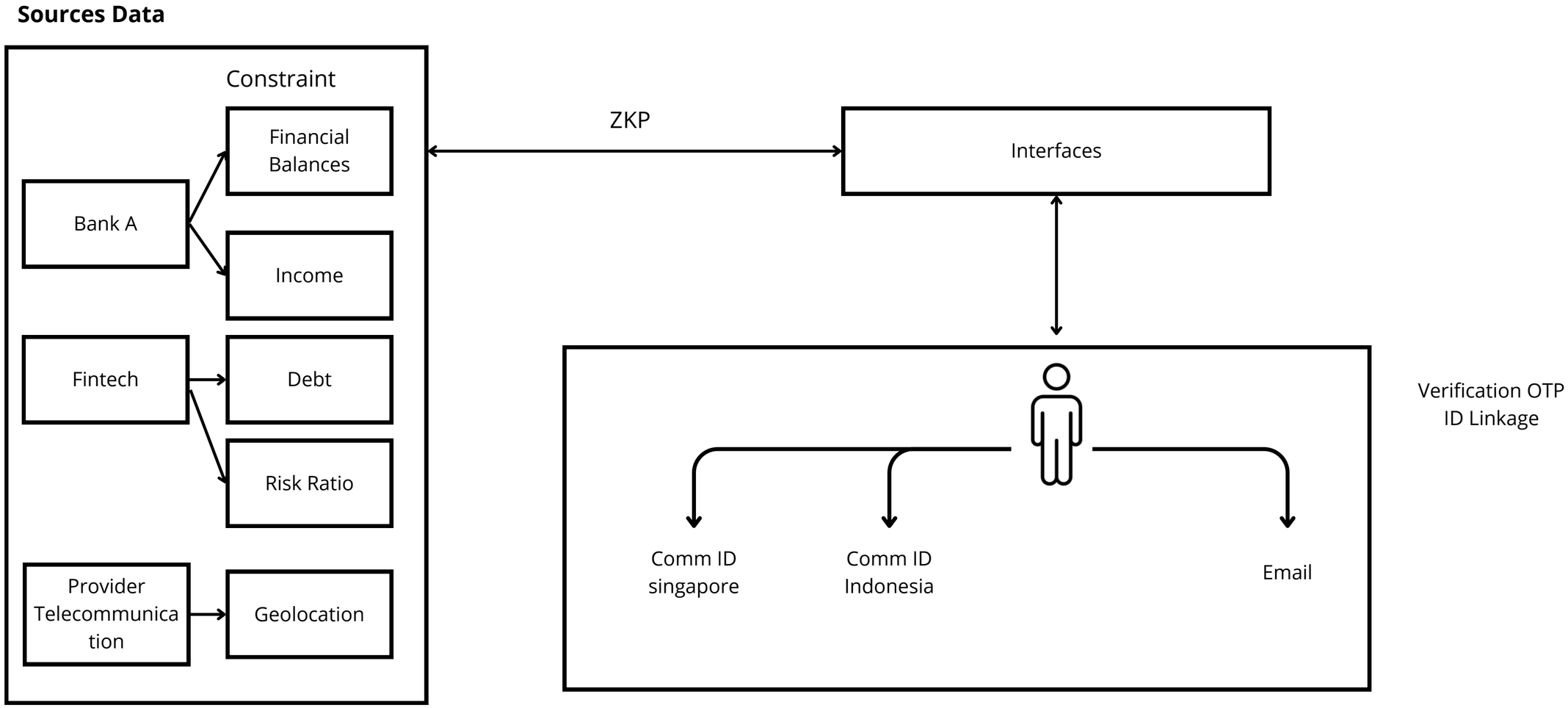


$$\Pi_{final} = H(\pi_A \parallel \pi_B \parallel \pi_C)$$

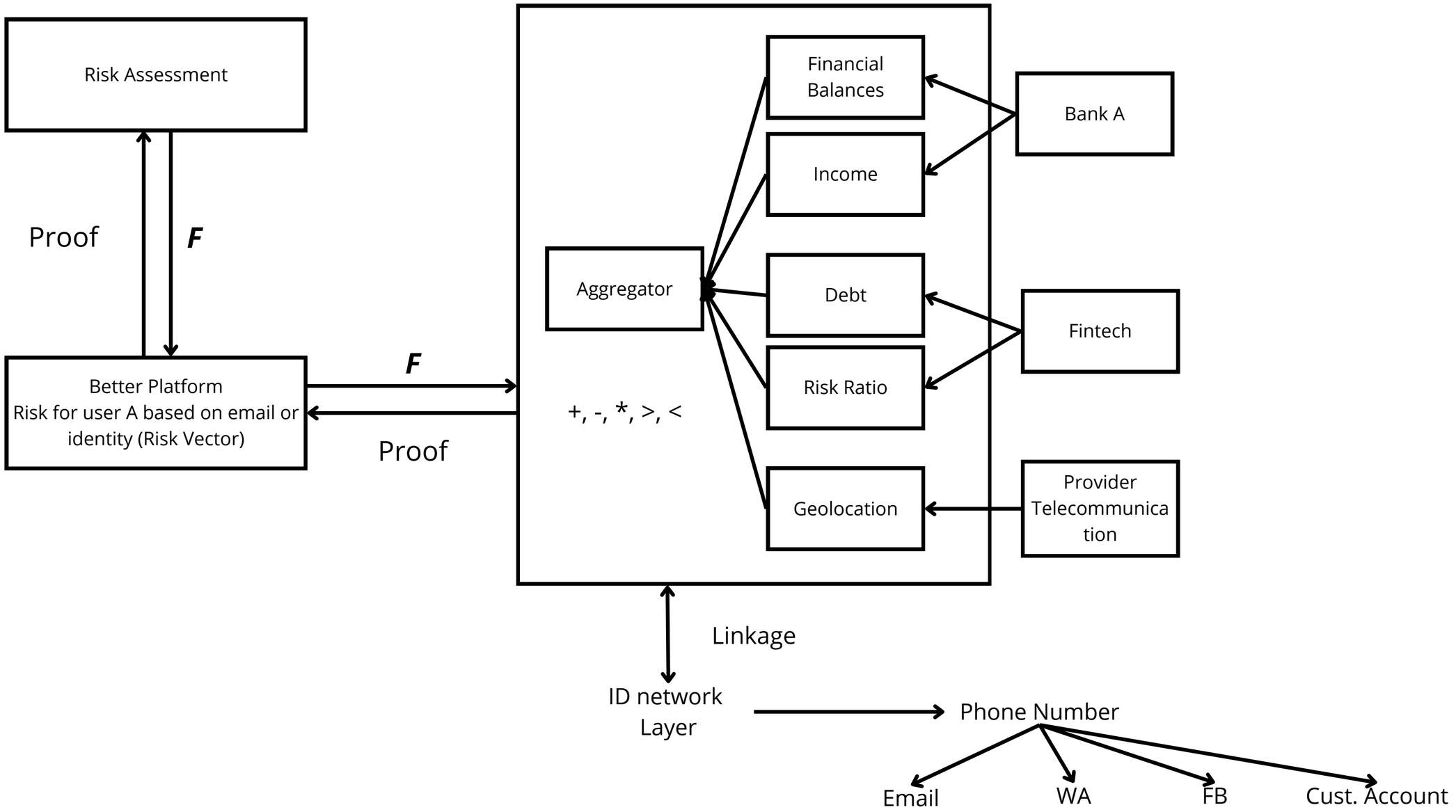
Single Prover:
All financial constraints (income, debt, risk ratio, payment history) are combined by a single prover to generate one monolithic proof.

Multi Prover:
Each institution independently generates its own sub-proof (e.g., Bank A → Proof A, Fintech → Proof B, E-commerce → Proof C).
All sub-proofs are securely aggregated into a final proof, ensuring distributed trust and privacy preservation.

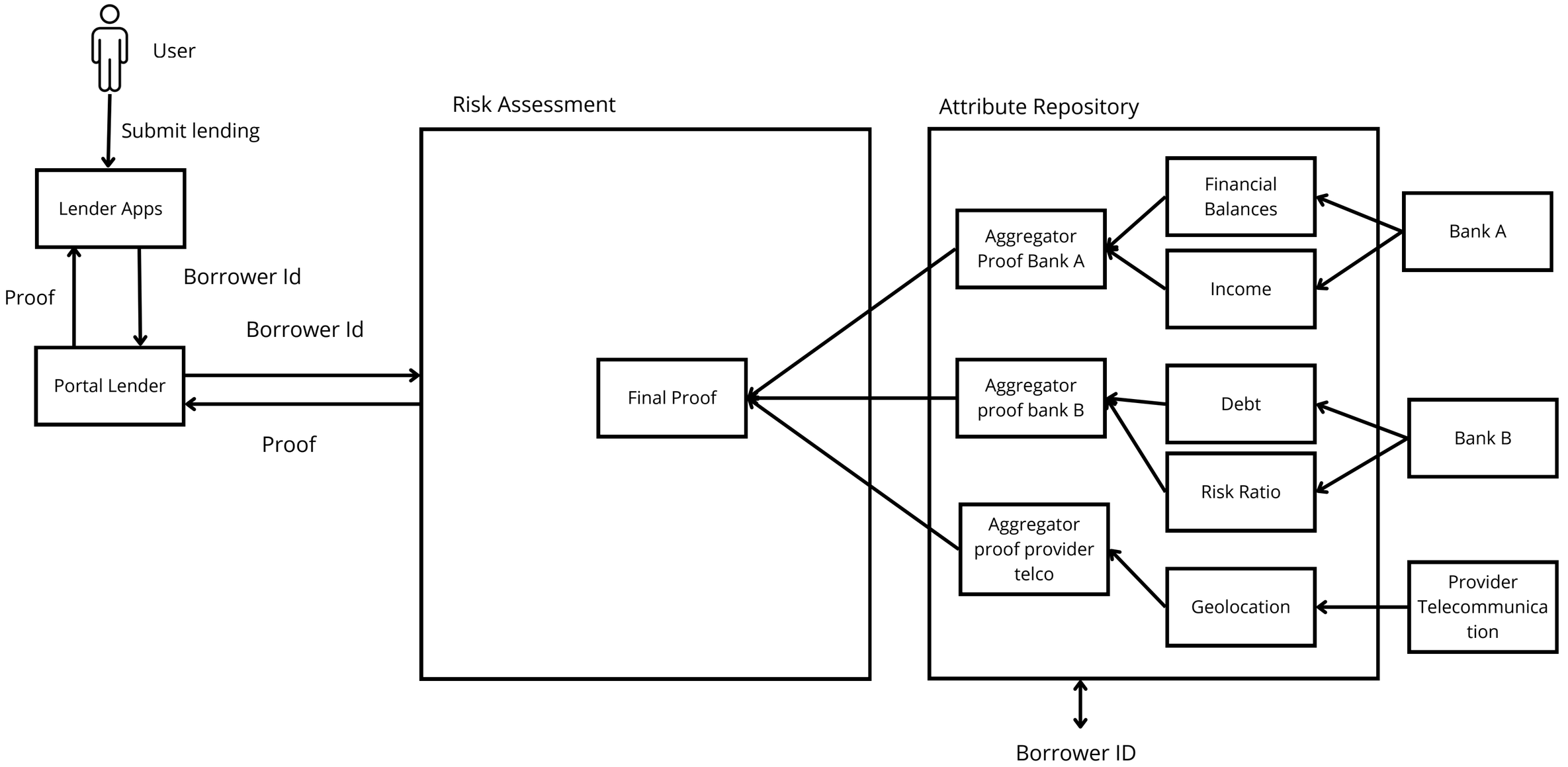
Composite prover ZK Multi NNP Architecture



Composite prover ZK Multi NNP Architecture



Composite prover ZK Multi NNP Architecture



DiStefano DCTLS (Designated Commitment Transport Layer Security)

Problem:

1. Informasi penting terdapat pada TLS seperti status pembayaran, usia, informasi akun, dan data pribadi lainnya
2. Data pada TLS hanya bisa diakses oleh client dan server.
3. Agar verifier dapat melakukan verifikasi maka data pribadi tersebut harus dibuka ke semua pihak sehingga melanggar data privacy.

Solusi lama:

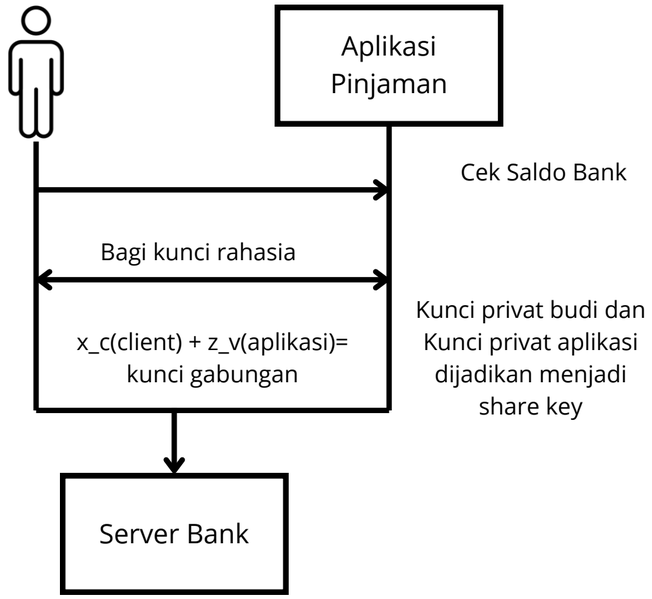
1. Client screenshot Verification → Client melakukan screenshot verification dan upload bukti verifikasi
2. Share session → Thirdparty atau verifier bisa akses seluruh data
3. Plaintext diberikan ke pihak ketiga
4. Menggunakan extension untuk verify
5. Meminta tambahan API dari server side

Solusi DCTLS:

1. Verifier dapat membuktikan user memiliki saldo > 10 juta tanpa memberikan data saldo pada server bank
2. Verifier dapat memverifikasi tanpa membuka plaintext, tanpa membuat API, tanpa menggunakan extension, tanpa share session sehingga data pribadi tidak diketahui oleh verifier.

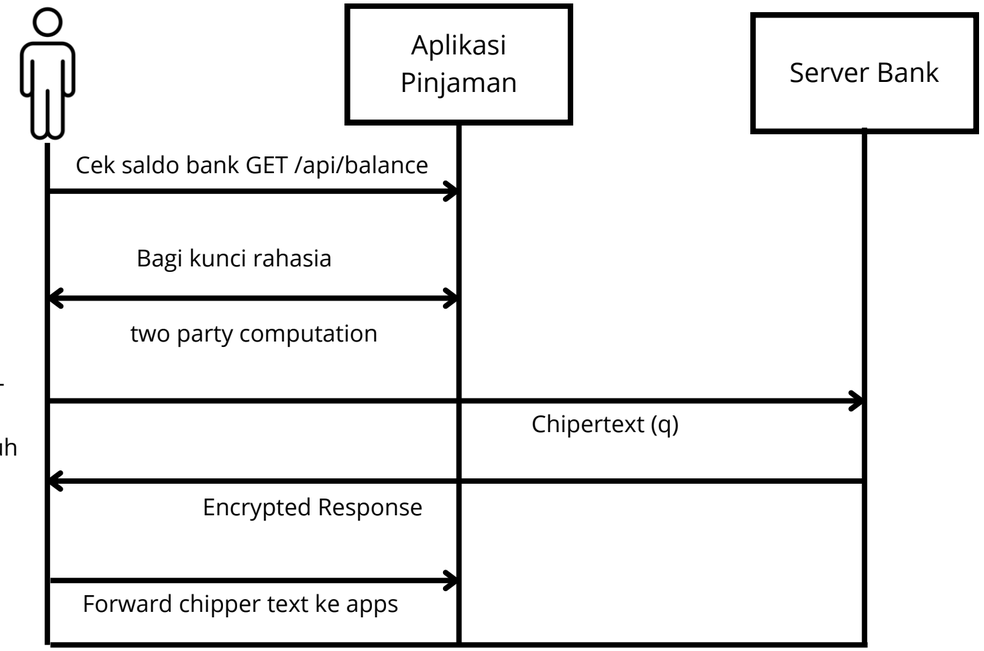
DiStefano DCTLS (Designated Commitment Transport Layer Security)

1. **Handshake**



- Client punya: tk_c (traffic key client) -
- Apps punya: tk_v (traffic key verify)
- Gabungan = tk_c + tk_v = kunci penuh

2. Query Phase



Input:
- Client: "GET /api/balance"
- Client: tk_c (bagian kunci)
- Apps: tk_v (bagian kunci

Proses:
Plaintext: "GET /api/balance" → AES-GCM Encryption dengan tk_c + tk_v → Ciphertext: "xa82j9..." → ke Bank

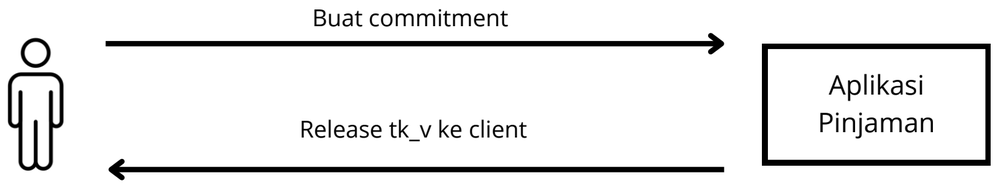
Output
RESPONSE dari BCA: Encrypted: "kj3h8f2..." (saldo tersembunyi)

Yang diterima
Client & Apps
r̂ = "encrypted(Saldo: Rp 15.000.000)"

Masih terenkripsi

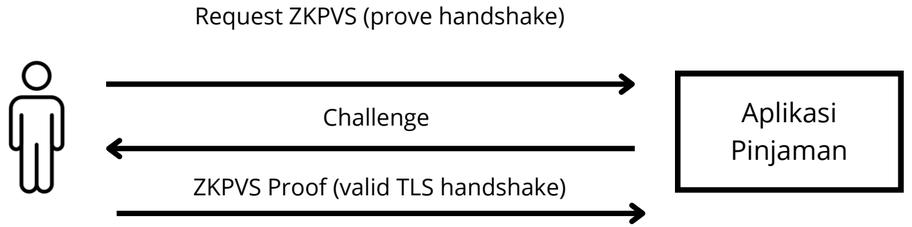
DiStefano DCTLS (Designated Commitment Transport Layer Security)

3. Commitment Phase



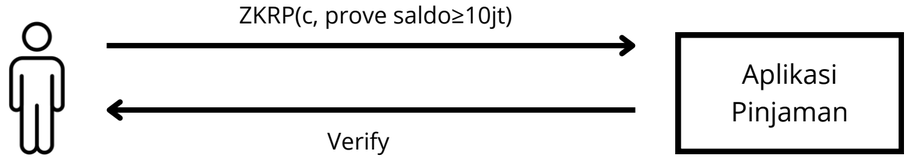
Fase dimana chipper text dari server bank sudah final dan tidak bisa diubah untuk memastikan data itu asli serta agar berlanjut ke tahap ZKRP.

4. ZKPVS



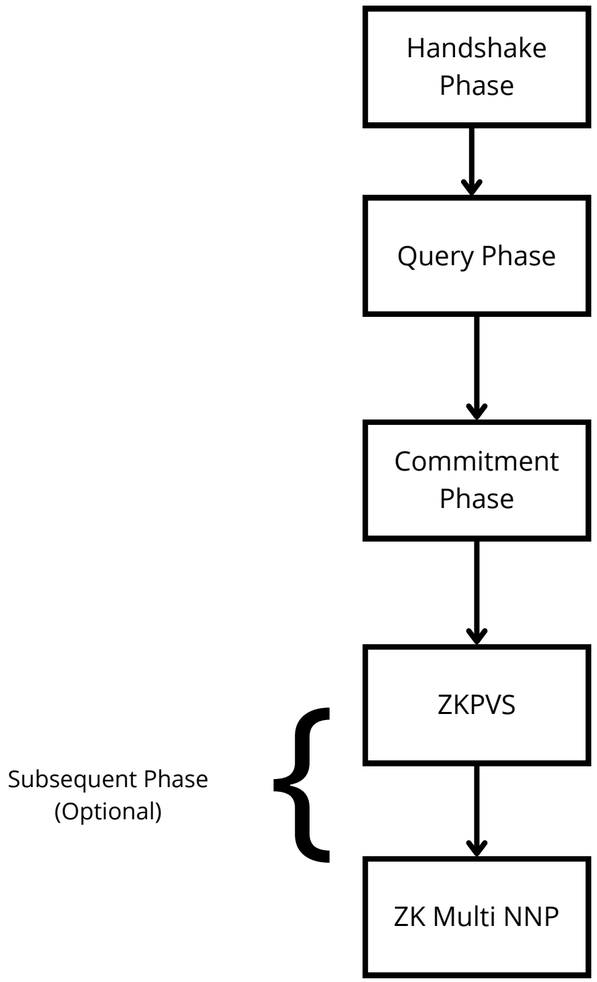
ZKPVS (Zero Knowledges Proof Valid Signature) agar aplikasi pinjaman dapat verifikasi dari bank terpercaya tanpa mengetahui nama banknya

5. ZKRP



DiStefano DCTLS (Designated Commitment Transport Layer Security)

1. **Pipeline**



1. Handshake Phase (2PC TLS 1.3)
Tahap awal untuk membangun sesi TLS secara aman dan privat menggunakan dua pihak (Client & Verifier).

- Kunci TLS dibagi menjadi dua share: tk_c (client) dan tk_v (verifier).
- Client dan Verifier bersama-sama melakukan proses handshake TLS tanpa mengungkapkan identitas server ke verifier.
- Menghasilkan session keys yang siap digunakan untuk enkripsi permintaan.

2**. Query Phase (2PC AES-GCM / Record Layer)**
Tahap pengiriman dan penerimaan data ke server menggunakan enkripsi gabungan (Client + Verifier).

- Client dan Verifier melakukan 2PC untuk mengenkripsi request, misalnya “GET /balance”.
- Ciphertext dikirim ke bank.
- Bank membalas ciphertext terenkripsi ($\hat{r}$).
- 2PC dekripsi memastikan hanya Client yang tahu plaintext, Verifier tetap buta terhadap data sensitif.

3. Commitment Phase
Tahap penguncian nilai ciphertext agar tidak bisa dimodifikasi oleh Client sebelum proses pembuktian.

- Client membuat komitmen: c = Commit($\hat{r}$).
- Client mengirim commitment c ke Verifier.
- Verifier yakin nilai yang akan dibuktikan adalah nilai asli dari server (integrity terjamin), tanpa melihat plaintext.

4. ZKPVS – Zero-Knowledge Proof of Valid Signature
Tahap pembuktian bahwa handshake TLS benar-benar berasal dari server yang sah, tanpa membuat Verifier mengetahui nama bank / identitas server.

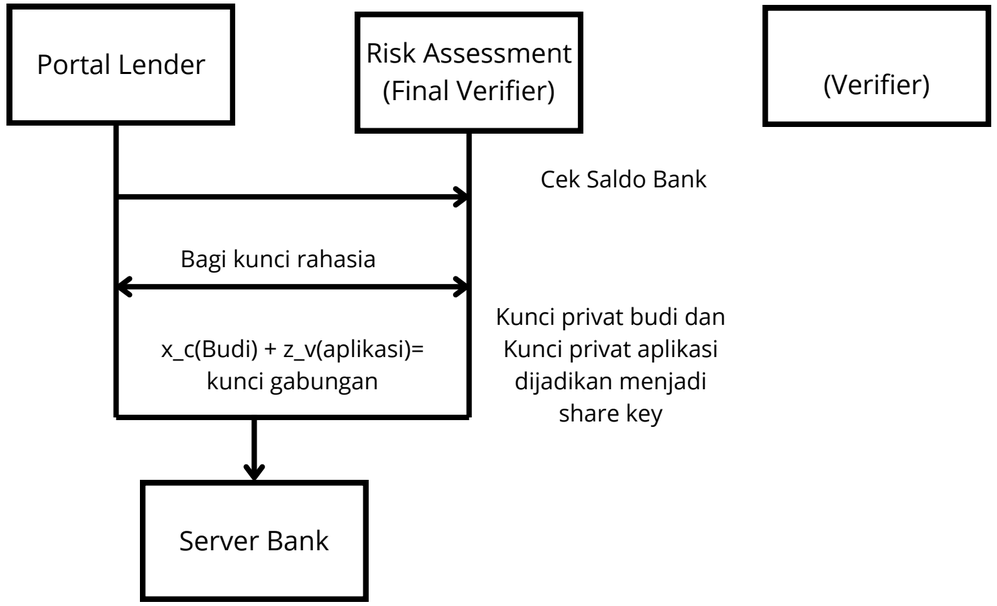
- Verifier mengirim challenge.
- Client memberikan ZK proof atas transcript handshake.
- Verifier yakin server valid, tetapi tetap tidak tahu server mana yang diakses oleh client.
- Menjamin authentication tanpa kehilangan privacy.

5. Zero Knowledges Multiplicative NNP
Tahap pembuktian sifat matematis dari saldo tanpa membuka saldo aslinya.

- Client membuktikan bahwa nilai dalam commitment memenuhi syarat, contoh: saldo ≥ 10 juta.
- Verifier hanya memeriksa proof, bukan angka saldo.
- Digunakan untuk pengambilan keputusan, misalnya approval limit pinjaman.

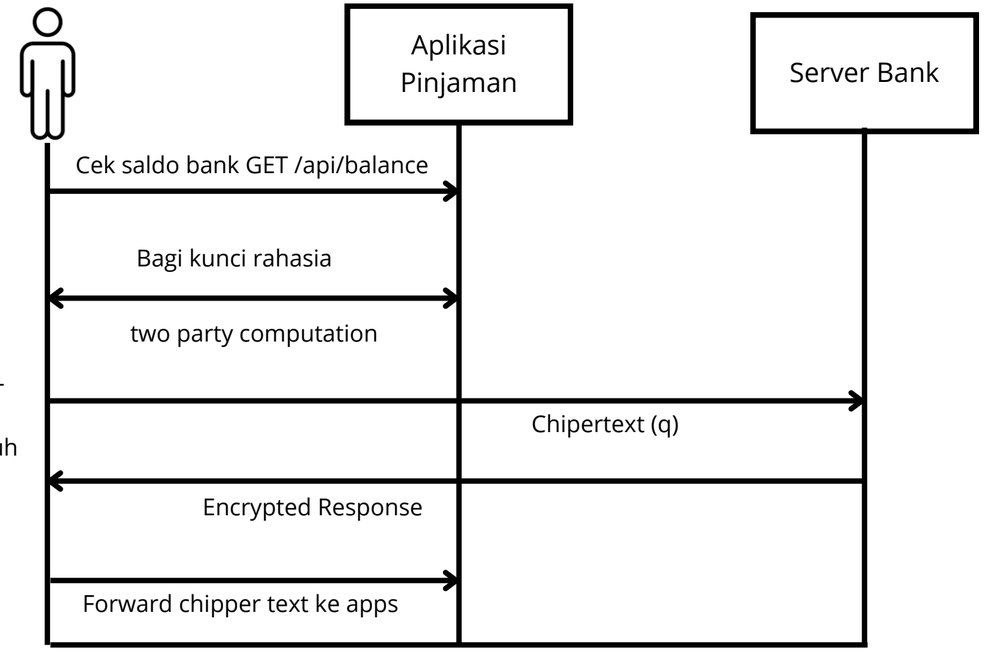
DiStefano DCTLS (Designated Commitment Transport Layer Security)

1. **Handshake**



- Client punya: tk_c (traffic key client) -
- Apps punya: tk_v (traffic key verify)
- Gabungan = tk_c + tk_v = kunci penuh

2. Query Phase



Input:
- Client: "GET /api/balance"
- Client: tk_c (bagian kunci)
- Apps: tk_v (bagian kunci

Proses:
Plaintext: "GET /api/balance" → AES-GCM Encryption dengan tk_c + tk_v → Ciphertext: "xa82j9..." → ke Bank

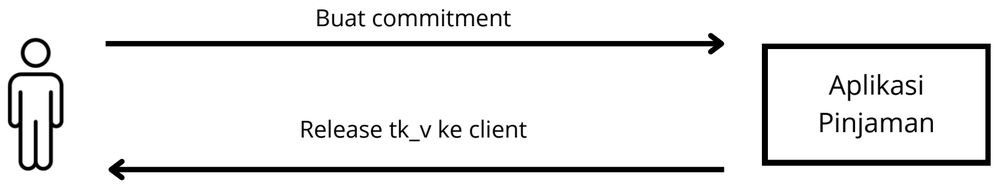
Output
RESPONSE dari BCA: Encrypted: "kj3h8f2..." (saldo tersembunyi)

Yang diterima
Client & Apps
r̂ = "encrypted(Saldo: Rp 15.000.000)"

Masih terenkripsi

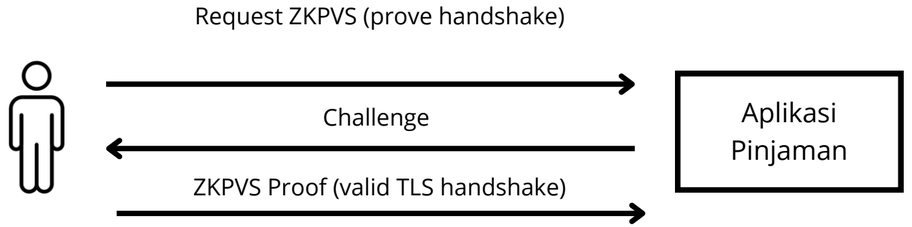
DiStefano DCTLS (Designated Commitment Transport Layer Security)

3. Commitment Phase



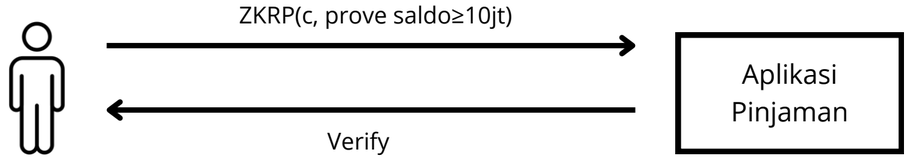
Fase dimana chipper text dari server bank sudah final dan tidak bisa diubah untuk memastikan data itu asli serta agar berlanjut ke tahap ZKRP.

4. ZKPVS

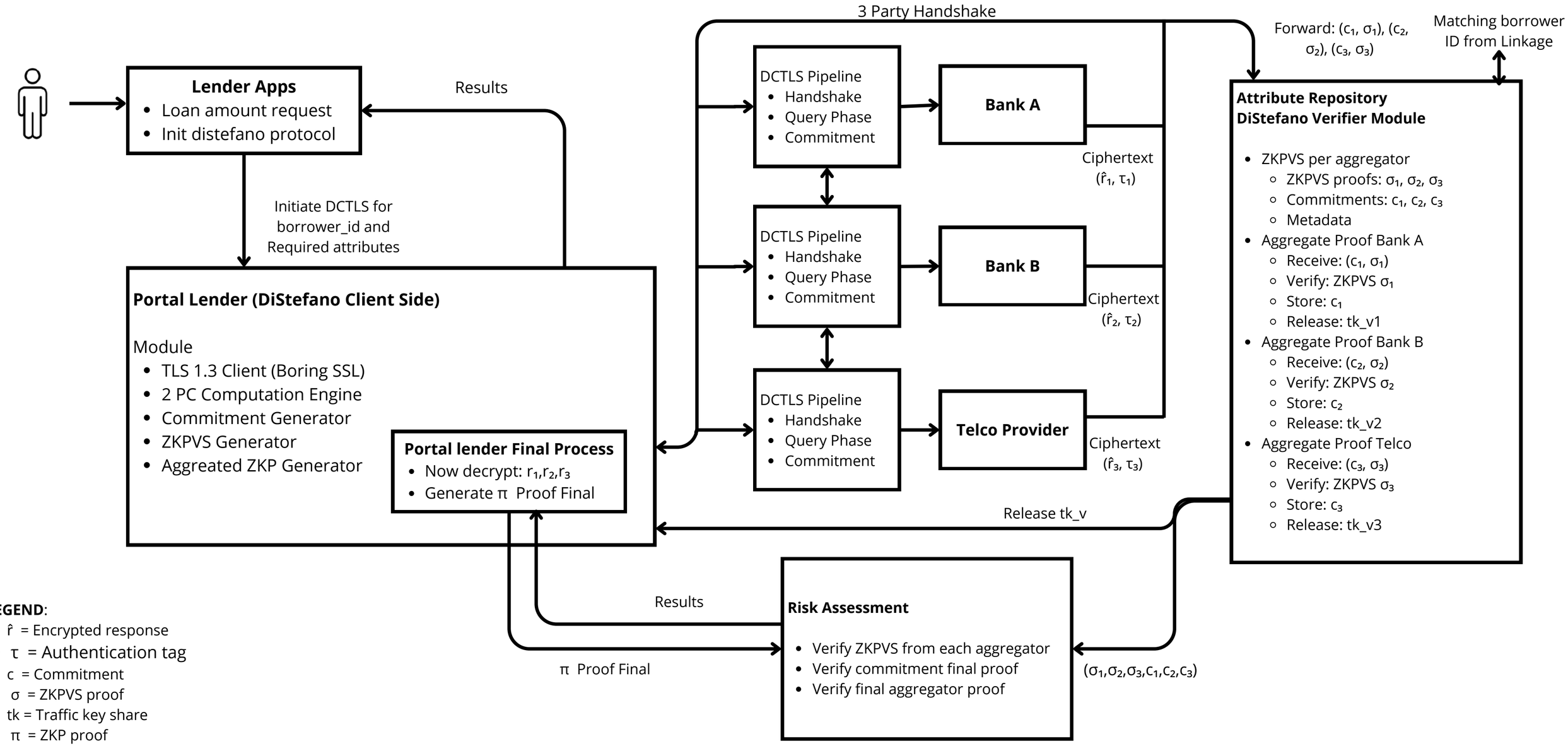


ZKPVS (Zero Knowledges Proof Valid Signature) agar aplikasi pinjaman dapat verifikasi dari bank terpercaya tanpa mengetahui nama banknya

5. ZKRP

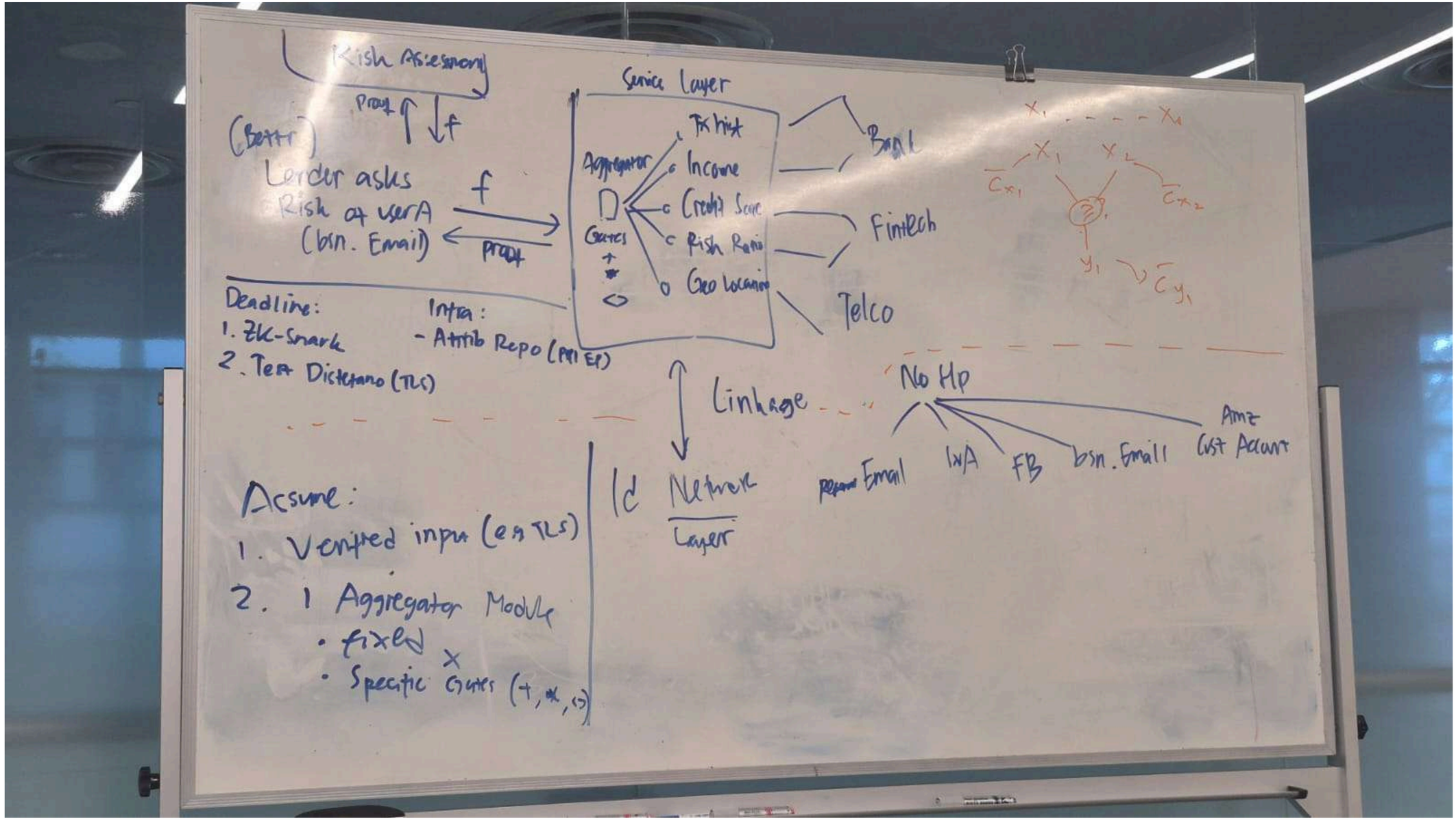


DiStefano DCTLS (Designated Commitment Transport Layer Security)



LEGEND:

- r̂ = Encrypted response
- τ = Authentication tag
- c = Commitment
- σ = ZKPVS proof
- tk = Traffic key share
- π = ZKP proof


Risk Assessment
Proof
f
(Better)
Lender asks
Risk of userA
(bsn. Email)
f
Proof
Service Layer
Aggregator
Gates
Tx hist
Income
Credit Score
Risk Ratio
Geo location
Bank
Fintech
Telco
Deadline:
1. ZK-Snark
2. Tee Distefano (TLS)
Intra:
- Attrib Repo (API EP)
Linkage
No Hp
Amz Cust Account
bsn. Email
FB
IxA
Email
Assume:
1. Verified input (e.g TLS)
2. 1 Aggregator Module
• fixed
• Specific Gates (+, *, <)
Id Network Layer